

Mathematics, 2013 EC, common, entrance exam

Grade 12 · Mathematics · 2013 E.C.

Time allowed: 120 minutes

Questions: 20

Total marks: 20

Name _____ Grade _____ ID No. _____ Date _____

1. What is the determinant of the matrix: (1)

$$\begin{pmatrix} 1 & -7 \\ 6 & 5 \end{pmatrix}$$

- A. 47 B. 37
C. -37 D. -47

2. The additive inverse of the complex number $3 - 4i$ is (1)

- A. $3 - 4i$ B. $-3 + 4i$
C. $3 + 4i$ D. $-3 - 4i$

3. What is the modulus of the complex number $-6 + 8i$? (1)

- A. 10 B. 8
C. 5 D. 2

4. Given two vectors $u = (1, 3)$ and $v = (-3, 5)$ in the plane. Then $6u + 2v$ is equal to (1)

- A. $(2, 28)$ B. $(0, 28)$
C. $(0, 18)$ D. $(3, 3)$

5. Which one of the following is the standard form of the equation of the circle centered at $(2, -3)$ and radius 5? (1)

- A. $(x - 2)^2 + (y + 3)^2 = 5$ B. $(x + 2)^2 + (y + 3)^2 = 25$
C. $(x - 2)^2 + (y + 3)^2 = 25$ D. $(x - 2)^2 + (y - 3)^2 = 25$

6. Which of the following defines the equation of a sphere whose center is at $(0, -1, 2)$ and radius 6 units? (1)

- A. $x^2 + (y + 1)^2 + (z - 2)^2 = 12$ B. $x^2 + (y + 1)^2 + (z + 2)^2 = 36$
C. $x^2 + (y + 1)^2 + (z - 2)^2 = 36$ D. $x^2 + (y - 1)^2 + (z - 2)^2 = 36$

7. Which of the following represents the vector $(3, -5, 7)$ in the space using the standard unit vectors i, j and k ? (1)

- A. $3i - 5j + 7k$ B. $-3i - 5j - 7k$
C. $3i - 5i - 7k$ D. $3i + 5j + 7k$

8. What are the domain and range of the function $f(x) = 2x^{3/4}$ respectively? (1)

- A. $[0, \infty)$ and $\mathbb{R}$ B. $[0, \infty)$ and $[0, \infty)$
C. $\mathbb{R}$ and $[0, \infty)$ D. $[0, \infty)$ and $(0, \infty)$

9. Which one of the following defines a one-to-one function? (1)

- A. $f(x) = \{(x, y): y = x^2 + 1\}$ B. $f(x) = \{(x, y): y \text{ is a father of } x\}$
C. $f(x) = \{(x, y): y \text{ is a sister of } x\}$ D. $f(x) = \{(x, y): y = 3x - 1\}$

10. The equation of a circle is given by $(x - 3)^2 + (y + 2)^2 = 9$. What are the center and the radius of the circle representatively? (1)
- A. (3,-2) and 9
B. (3,2) and 9
C. (3,-2) and 3
D. (3,2) and 3
-
11. Which of the following sentences has a truth value? (1)
- A. Man is mortal
B. Tomorrow is a beautiful day
C. Man is selfish
D. We shall win!
-
12. Which one of the following is the standard form of the equation of the parabola with vertex (4, 6) and focus (-8, 6)? (1)
- A. $(y - 6)^2 = 48(x - 4)$
B. $(y - 6)^2 = -48(x - 4)$
C. $(x - 4)^2 = -12(y - 6)$
D. $(x - 4)^2 = 12(y - 6)$
-
13. Which of the following is true about the hyperbola given by $(x^2/4) - y^2 = 1$? (1)
- A. Its center is at (2, 1).
B. An asymptote to the hyperbola is $y = 2x$
C. Its foci are at (-5, 0) and (5, 0).
D. Its vertices are at (-2, 0) and (2, 0).
-
14. Two statements P and q are given by: (1)
- p: $22/7$ is an irrational number.
q: The sum of two odd integers is odd.
- Then which of the following has truth value T?
- A. $\neg p \wedge q$
B. $p \wedge q$
C. $p \wedge \neg q$
D. $\neg q \vee p$
-
15. Which one of the following is the contrapositive of the statement: If 21 is a prime number, then 4 is an odd number? (1)
- A. If 4 is not an odd number, then 21 is not a prime number.
B. If 21 is not a prime number, then 4 is an even number.
C. If 4 is an odd number, then 21 is not a prime number.
D. If 4 is an odd number, then 21 is a prime number.
-
16. In a certain health center there are 3 doctors, 8 nurses and 2 physicians. In how many ways one can form a group of 5 members consisting of 1 doctor, 3 nurses and 1 physician? (1)
- A. 336
B. 330
C. 168
D. 112
-
17. Which one of the following forms represents an exhaustive set of events if a die is thrown once? (1)
- A. {1, 2}, {2}, {4, 5}, {5, 6}
B. {1, 2}, {4}, {4, 5, 6}
C. {1, 2, 3}, {3, 4}, {6}
D. {1, 2, 3}, {4, 5}, {6}
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18. A bag contains 2 red, 3 yellow and 5 green balls. One ball is drawn at random. What is the probability that the drawn ball is NOT yellow? (1)
- A. $4/5$
B. $7/10$
C. $3/10$
D. $1/5$

19. Which one of the following is NOT true about the determinant of a given square matrix A ? (1)

- A. The determinant of A is equal to the determinant of its transpose.
- B. Interchanging two rows of A gives the same determinant.
- C. If A contains a row of zeros, then its determinant is zero.
- D. If A has identical rows, then its determinant is zero.

20. If f is a continuous function on $[a, b]$ and F is its anti-derivative, then which of the following is true? (1)



Question Diagram

- | | |
|------|------|
| A. A | B. B |
| C. C | D. D |

— END OF EXAMINATION —

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